

Janus Doping of Sulfur into Platinum Diselenide Ribbons

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2D platinum diselenide (PtSe_2), a novel member of the transition metal dichalcogenides (TMDCs) family, possesses many excellent properties, including a layer-dependent bandgap, high carrier mobility, and broadband response, making it promise for applications in technologies like field-effect transistors and room-temperature photodetectors. Doping represents an effective method to modify the electrical properties of 2D TMDCs and to bestow upon them additional functions. However, to date, little research has been conducted on the successful doping of 2D PtSe_2 for modification. In this study, sulfur (S) powder is utilized during the chemical vapor deposition growth process of 2D PtSe_2 ribbons and successfully integrated into the PtSe_2 lattice through substitutional doping. The Au substrate significantly decreases the substitution energy of Se atoms in the lower layer of PtSe_2 , resulting in the formation of the Janus PtSSe structure. S-doped PtSe_2 ribbons demonstrate significant symmetry breaking and enhanced electrical properties, showcasing a strong nonlinear optical response and certain synaptic plasticity, further simulating some neuromorphological processes. This study not only demonstrates a viable method for controllable doping and modification of 2D PtSe_2 but also establishes a platform for exploring the characteristics of Janus TMDCs.

Semiconducting PtSe_2 is a promising candidate for 2D electronic and optoelectronic devices, while semi metallic PtSe_2 may offer an important platform for exploring novel quantum phenomena.^[2] To date, significant progress has been made in the growth of 2D PtSe_2 using methods such as chemical vapor deposition (CVD),^[2,3] chemical vapor transport,^[4] thermally assisted conversion,^[5] molecular beam epitaxy,^[6] and mechanical exfoliation.^[7] However, for optimal performance, the physical properties of 2D PtSe_2 require modification. Doping serves as an effective method for adjusting material properties and device performance, encompassing strategies such as intercalation, surface charge transfer doping, and substitutional doping. Moreover, substitutional doping is recognized as the most stable and non-volatile method.^[8] By incorporating diverse elements into the lattice of 2D transition metal dichalcogenides (TMDCs), it is possible to effectively expand their

functions, thereby offering a novel strategy for contact engineering and carrier modulation in electronic devices based on 2D TMDCs.^[9] For instance, the doping of vanadium (V) and iron atoms has endowed molybdenum disulfide (MoS_2) and tungsten disulfide (WS_2) with room-temperature ferromagnetism.^[10] MoS_2 when substituted by niobium atoms, exhibited p-type transport behavior.^[11] In semiconducting V-doped WSe_2 monolayers, room-temperature ferromagnetic domains exhibiting long-range order were identified.^[12] In addition, Janus TMDC, representing a new type of asymmetric TMDC material, has recently attracted widespread research interest.^[13] By replacing one layer of chalcogen X in MX_2 with another chalcogen Y, it is possible to form a Janus structure with the general formula MX_Y . Due to the disruption of planar symmetry, Janus MX_Y often exhibits unique electronic, structural, thermoelectric, and optical properties that differ significantly from those of 2D MX_2 .^[14] The substitution reaction of chalcogen elements is utilized to adjust the composition of TMDC and tailor bandgaps.^[15]

Over the past few years, CVD methods have achieved significant progress in the area of substitution doping of TMDCs. For instance, V atoms could enter the lattice of MoS_2 during CVD growth by decomposing solid V precursors, resulting in tunable doping concentrations across a wide range. The replaced V atoms were uniformly distributed in V-doped MoS_2 , yielding stable and uniform doping effects and electronic properties.^[9]

1. Introduction

The 2D layered atomic crystal platinum diselenide (PtSe_2) has garnered widespread interest due to its potential applications in electronics, optoelectronics, catalysis, and magnetoelectronic.^[1] This layered PtSe_2 can undergo a transition from a semiconductor to a semimetal as the number of layers increases.

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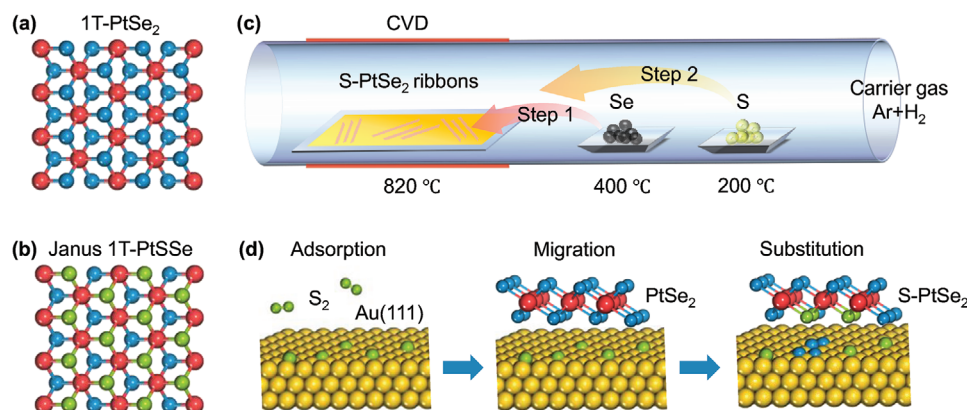


Figure 1. Top view of atomic structure of a) PtSe_2 ; b) Janus PtSSe . c) Schematic diagram of the CVD synthesis of $\text{PtS}_x\text{Se}_{2-x}$ ribbons. d) Schematic diagram of the growth mechanism of $\text{PtS}_x\text{Se}_{2-x}$ ribbons.

Rhenium (Re) and V atoms could be substituted in MoS_2 , WS_2 , and WSe_2 by spin coating precursor solutions onto the substrate. Tunable composition, optical, and electrical properties could be achieved in Re- and V-doped TMDC monolayers.^[16] Re atoms could also be introduced into the lattice of MoS_2 through metal-organic CVD. The transmission characteristics of Re-doped MoS_2 back-gated field-effect transistors (FETs) show significant improvement compared to undoped MoS_2 .^[8] However, due to the good stability and strong interlayer coupling,^[17] current research on successfully doping 2D PtSe_2 to achieve modification is limited, although theoretical calculations suggested^[18] that 2D PtSe_2 could be modified by substitutional doping.

In this study, sulfur (S) powder served as a precursor in the CVD growth of 2D PtSe_2 ribbons, facilitating the Janus doping of S atoms into 2D PtSe_2 . According to density functional theory (DFT) calculations, Janus doping was attributed to the decrease in energy barriers during the adsorption, migration, and substitution processes of S atoms on the Au substrate. The doped ribbons demonstrated obvious symmetry breaking and significant changes in electrical properties, including the ON-OFF ratio and carrier mobility. This study presents a feasible strategy for the controllable doping and modification of 2D PtSe_2 and establishes a platform for studying the properties of Janus PtSSe and TMDCs doped with non-metallic atoms.

2. Results and Discussion

2.1. CVD Growth of S-Doped PtSe_2 ($\text{PtS}_x\text{Se}_{2-x}$) Ribbons

Figure 1a,b depict top views of the atomic structures of 1T- PtSe_2 and Janus 1T- PtSSe , respectively. In the top view of 1T- PtSe_2 , each Pt atom (red) is surrounded by six Se atoms (blue). The side view illustrates Pt atoms in the central layer connected to three Se atoms in both upper and lower layers (Figure S1a, Supporting Information). Complete substitution of the upper or lower Se atomic layers with S atomic layers results in Janus 1T- PtSSe (Figure S1b, Supporting Information), where each Pt atom in the top view is surrounded by three Se atoms and three S atoms (green).

An atmospheric pressure CVD method was previously developed for synthesizing 2D 1T- PtSe_2 ribbons, exhibiting high crys-

tal quality and high electrocatalytic hydrogen evolution activity.^[3b] To modify their physical properties and further enhance functionality, the synthesis method was refined by placing S powder as a dopant upstream of the Se powder, resulting in $\text{PtS}_x\text{Se}_{2-x}$ ribbons (Figure 1c). The preparation process comprised two steps: 1) Employing the previously procedure, utilize 10 μL $(\text{NH}_4)_2\text{PtCl}_4$ aqueous solution with a concentration of 0.1–1 mg mL^{-1} and 800 mg of Se powder as precursors, reacting at 820 °C for 10 min to fabricate monolayer PtSe_2 ribbons. 2) Maintaining the temperature of the heating zone constant, S powder was heated to gradually evaporate and diffuse into the reaction zone via the carrier gas, thereby the PtSe_2 ribbons were doped for 5 min. The flow rate of the precursors was closely related to the quality of the products, and reducing the feed rate of the precursors can reduce nucleation density, thus leading to large crystal domains.^[17] In this study, the temperature of the S powder was maintained at 200 °C to regulate the evaporation rate. It was noteworthy that the distance between the S powder and Se powder must be precisely maintained at 5 cm. When the distance was too close, the reaction between the S powder and Pt precursors became dominant, potentially producing more by-products like PtS_2 . The morphologies of PtS_2 included irregular polygons or ribbons (Figure S2a–d, Supporting Information), enabling determination of product purity based on these morphologies. When the distance was too far, the doping effect of S atoms became poor, resulting in products that were primarily PtSe_2 ribbons.

The Au (111) substrate could play a crucial role in the Janus doping process, frequently resulting in the dissociation, chemical adsorption, and migration of gaseous S molecules. Through adsorption and migration on Au substrate, S atoms could substitute the Se layer at the bottom of MoSe_2 crystals, resulting in the formation of Janus MoSSe .^[19] To delve deeper into the formation mechanism of the Janus PtSSe region, DFT calculations were conducted to assess the pertinent energies of the CVD reaction process (Figures S3–S5, Supporting Information). Monolayer PtSe_2 ribbon and Au (111) supercells were constructed as the research system, which aligned with the theoretical values of the material's cell parameters. The computed adsorption energies of S atoms on the PtSe_2 ribbon and Au (111) surface were -1.37 and -3.79 eV, respectively, suggesting that S atom adsorption on the Au (111) surface was energetically more favorable.

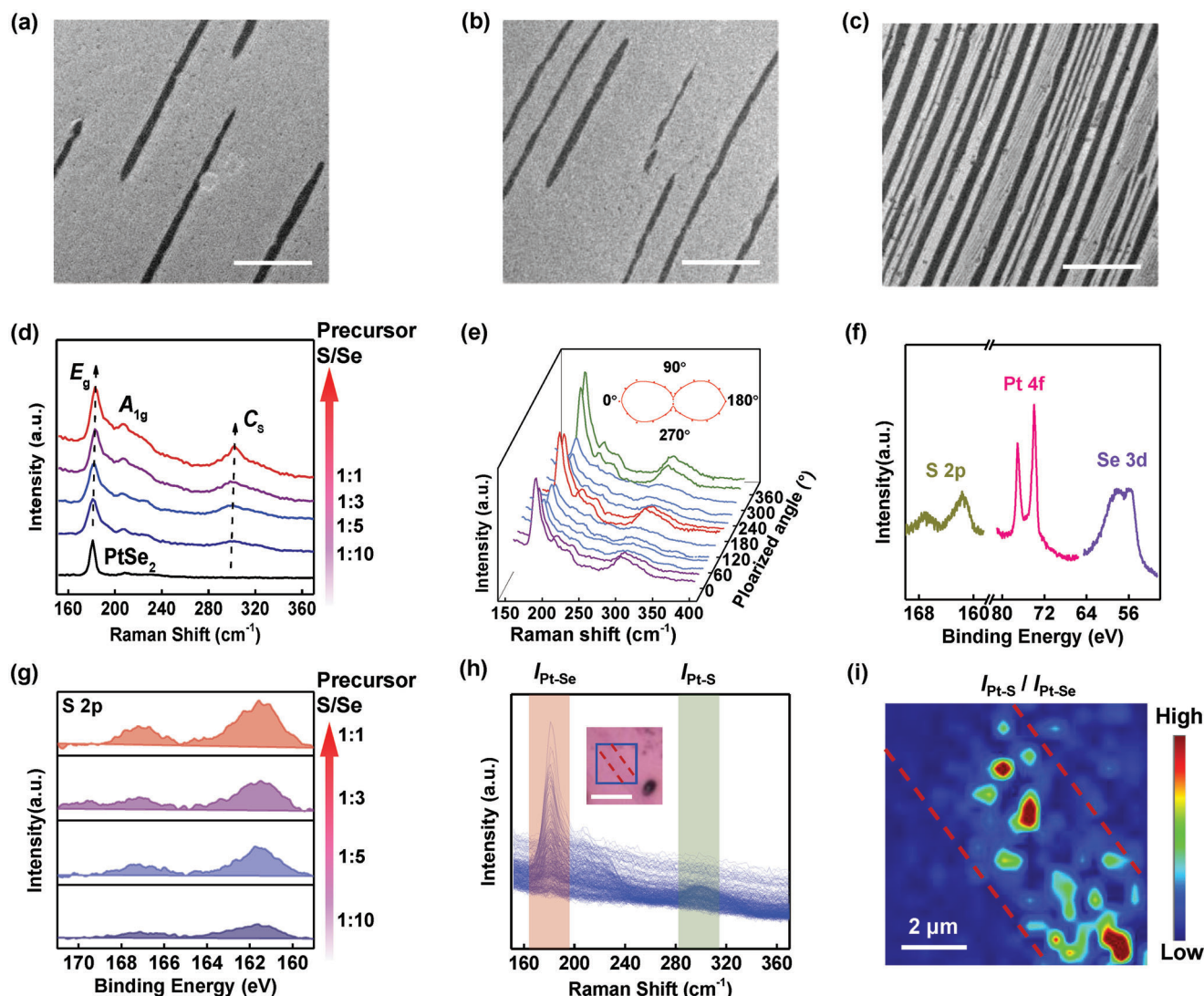


Figure 2. Scanning electron microscope (SEM) images of $\text{PtS}_x\text{Se}_{2-x}$ ribbons synthesized at different precursor concentrations: a) 0.1 mg mL^{-1} ; b) 0.5 mg mL^{-1} ; c) 1 mg mL^{-1} (Scale bars: $5 \mu\text{m}$). d) Raman spectra of $\text{PtS}_x\text{Se}_{2-x}$ ribbons synthesized at different precursor S/Se ratio, from 1:10 to 1:1. e) Polarized Raman spectra of $\text{PtS}_x\text{Se}_{2-x}$ ribbons synthesized at a precursor ratio of 1:1, which were collected when the rotation angle θ was set from 0° to 360° . Inset is the polar plot of the C_s peak intensity as a function of the rotation angle θ . f) XPS spectra of $\text{PtS}_x\text{Se}_{2-x}$ ribbons synthesized at a precursor ratio of 1:1. g) XPS spectra of $\text{PtS}_x\text{Se}_{2-x}$ ribbons synthesized at different S/Se ratios. h) Surface scan Raman spectra of the $20 \times 20 \mu\text{m}^2$ area. Inset is the optical microscope (OM) image of the scanning area (Scale bar: $10 \mu\text{m}$). i) Distribution map of peak area ratio between C_s peak and E_g peak.

The Nudged Elastic Band (NEB) curve illustrated that the migration barrier of S and Se atoms in the neighboring octahedral gaps on the Au (111) surface was low ($\approx 0.5 \text{ eV}$), and they exhibited high migration rates at elevated temperatures during the CVD reaction. The substitution energy of S atoms replacing the upper and lower Se atoms in the PtSe_2 ribbon were calculated as 3.14 and 0.67 eV for the upper and lower Se atom substitutions, respectively, which indicated that the energy required for S atoms to replace the upper Se atoms in the PtSe_2 band was notably higher than that for replacing the lower Se atoms. Based on the calculation results, the growth mechanism of $\text{PtS}_x\text{Se}_{2-x}$ ribbons could be analyzed, which primarily comprised three processes: adsorption, migration, and substitution (Figure 1d). At high temperatures, S_2 molecules decomposed into S atoms, which then ad-

sorbed onto the surface of Au substrates. These S atoms subsequently migrated to the vicinity of the PtSe_2 ribbons on the surface of Au substrates, replacing the lower Se atoms and forming the Janus PtSSe region.

2.2. Characterizations of $\text{PtS}_x\text{Se}_{2-x}$ Ribbons

$\text{PtS}_x\text{Se}_{2-x}$ maintained ribbon morphologies, analogous to the original PtSe_2 ribbons (Figure 2a–c). The size and distribution density of $\text{PtS}_x\text{Se}_{2-x}$ ribbons were positively correlated with the concentration of metal precursors, akin to PtSe_2 ribbons. Figure S6 (Supporting Information) illustrated a typical atomic force microscope (AFM) image of the $\text{PtS}_x\text{Se}_{2-x}$ ribbons, revealing a

measured band thickness of 0.92 nm, maintaining a monolayer structure. Raman spectroscopy serves as a convenient method to distinguish between doped and undoped samples, and to investigate how S atoms affect the structure and vibrational modes of PtSe₂ ribbons. Figure 2d presents the Raman spectra of pristine PtSe₂ ribbons and doped products obtained at various precursor ratios of S/Se, highlighting significant differences between doped and undoped samples. The Raman spectrum of pristine PtSe₂ ribbons featured two characteristic peaks: the E_g peak, indicative of the in-plane vibration mode (181.5 cm⁻¹), and the A_{1g} peak, representative of the in-plane vibration mode (208.5 cm⁻¹). As the precursor ratio of S/Se increased, the E_g peak of the PtS_xSe_{2-x} exhibited a slight blue shift of ≈ 1 cm⁻¹ with a larger peak width, and this shift amplified, suggesting that the introduction of S altered the in-plane structure and charge distribution. Additionally, a distinct peak (C_s) emerged at 302.0 cm⁻¹, sensitive to the S/Se ratios. This peak's intensity increased with a higher proportion of S, a phenomenon also observed in other doped TMDCs.^[9] Interestingly, the intensity of the C_s peak fluctuated with the rotation angle of the half-wave plate in the Raman instrument (θ) (Figure 2e). When θ was set to 0° and 180°, the C_s peak intensity reached its maximum value, whereas at 90° and 270°, it diminished, indicating that the C_s peak intensity was sensitive to the polarization angle of the incident light, which was a distinctive characteristic distinguishing doped from undoped samples (inset of Figure 2e). In the X-ray photoelectron spectroscopy (XPS) analysis of PtS_xSe_{2-x}, signals of substituted S atoms appeared at 167.4 eV ($2p_{1/2}$) and 161.6 eV ($2p_{3/2}$), suggesting an influence of S atoms on the chemical properties of PtSe₂ (Figure 2f). Furthermore, the peak area of the S signals in XPS could be utilized to calculate the content of the S element within the entire ribbon. As the S/Se ratio in the precursors was increased from 1:10 to 1:1, the proportion of S atoms rose to 4.3%, 10.2%, 11.0%, and 19.7%, respectively (Figure 2g). A 10 × 10 μm² area (inset of Figure 2h) with a step size of 0.5 μm was selected, and Raman surface scanning conducted on the monolayer PtS_xSe_{2-x} ribbon yielded a Raman spectrum containing 400 sampling points, as illustrated in Figure 2h. Using the ratio of the peak areas of C_s and E_g peaks as the indicator, a ratio distribution map consistent with the actual position of the ribbon was created (Figure 2i). The red area indicated a higher C_s peak intensity, correlating to a higher doping concentration, whereas the blue indicated a higher C_s peak intensity, correlating with a higher doping concentration, while the blue area represented a weaker C_s peak and consequently, a lower doping concentration. The uneven intensity distribution of C_s peaks within the ribbon suggested that S doping occurred randomly.

2.3. Crystal Structure and Second Harmonic Generation (SHG) of PtS_xSe_{2-x} Ribbons

A high-angle annular dark field scanning transmission electron microscope (HAADF-STEM) was employed to study the atomic structure of monolayer PtS_xSe_{2-x} ribbons. Within the ribbons, two distinct lattice structures were observed: 1T PtSe₂ (Figure 3a), characterized by the upper and lower layers of each Pt atom bonded to three Se atoms (Figure 1a), and the Janus PtSSe structure (Figure 3b), which consisted of a Pt layer sandwiched be-

tween two different S and Se layers (Figure 1b). S atoms and Se atoms could be directly observed and distinguished in HAADF-STEM images due to their dimmer contrast, stemming from the image contrast being related to the atomic number of the imaged species (Figure 3c,d). Consequently, S atoms selectively replaced regions of pristine PtSe₂ ribbons. Based on the monolayer characteristics, atomic structure, and HAADF-STEM images of PtS_xSe_{2-x} ribbons, the formation of ordered Janus PtSSe areas has been confirmed. In addition to these regions, a small number of Se vacancies and suspended Se atoms were also identified (Figure S7, Supporting Information).

Janus TMDCs can break the mirror symmetry and generate planar polarity due to charge imbalance.^[20] The breaking of mirror symmetry can lead to physical effects and functionalities beyond those of traditional TMDCs, particularly in nonlinear optics, yielding powerful nonlinear responses.^[21] As a nonlinear optical process, SHG can convert two photons of the fundamental frequency ω into a single photon of the harmonic frequency 2ω ,^[22,23] and is typically employed to determine lattice orientation, detect symmetry breaking, identify edge states in TMDCs^[24] and analyze grains in polycrystalline films.^[25] Ideal single crystal PtSe₂ is categorized within the D_{3d} point group, possessing a central symmetric structure, and exhibits no SHG effect. Conversely, Janus PtSSe monolayers belong to the C_{3v} point group, owing to their lack of mirror symmetry, and exhibit strong SHG signals.^[26] SHG signals from monolayer PtS_xSe_{2-x} and PtSe₂ were utilized to explore the nonlinear characteristics of as-grown 2D crystals using back-reflection geometry, further enabling a comparison of the degree of symmetry breaking between doped and undoped products. The excitation source consisted of a femtosecond optical parametric oscillator laser with a repetition rate of 76–80 MHz. The laser, emitting at a wavelength of 800 nm, was nearly perpendicularly incident on the sample, with its doubling spectrum at 400 nm being detected (Figure 4a). Polarization-dependent SHG in both monolayer PtSe₂ (Figure 4b) and PtS_xSe_{2-x} (Figure 4c) ribbons exhibited six-fold rotational symmetry, but the intensity in PtSe₂ (≈ 6000 cps) was significantly lower than that in PtS_xSe_{2-x} ($\approx 150\,000$ cps). For crystal structures in the C_{3v} point group, the SHG intensity variation is described by the equation $I = I_0 \cos^2 3\theta$, where θ is the azimuth angle between the polarization of the incident laser and the armchair direction, with I and I_0 denoting the SHG intensity and its maximum response, respectively.^[27] Thus, in the Janus PtSSe region with broken inversion symmetry, the SHG intensity theoretically reached its maximum value when the polarization direction aligns with one of the six armchair directions (Figure 4d), yielding a six-fold symmetric anisotropic response. As previously mentioned, perfect PtSe₂ lattices do not generate second harmonics, with the lower intensity in PtSe₂ potentially resulting from the interfacial interaction due to the Au substrate and symmetry breaking due to defects at the edges and irregularly arranged atoms.^[3b] This significant enhancement of the SHG effect in PtS_xSe_{2-x} could be attributed to the introduction of non-centrosymmetric structures by Janus PtSSe regions. Furthermore, surface scan tests of monolayer PtSe₂ (Figure 4e) and PtS_xSe_{2-x} (Figure 4f) ribbons revealed that the SHG intensity distribution within a 10 × 10 μm² area mirrored the actual ribbon shape. The irregular SHG intensity distribution verified the random distribution of the doping regions, indicating that S atoms were locally concentrated within the

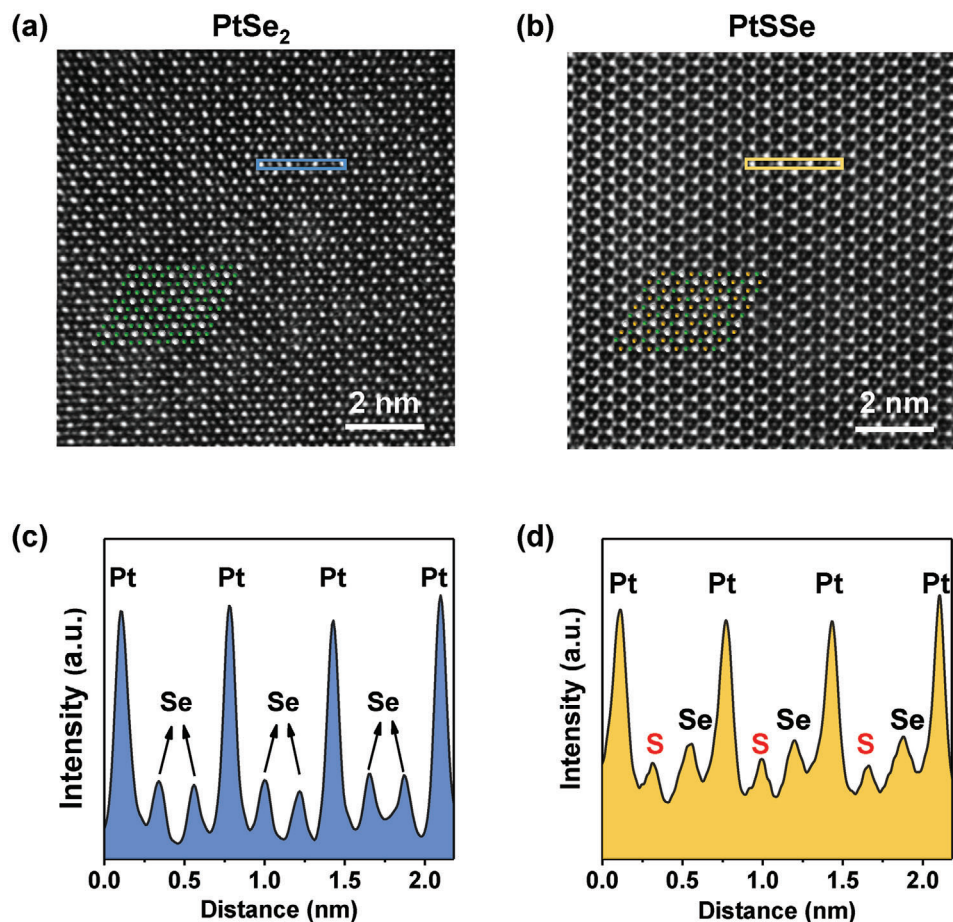


Figure 3. HAADF-STEM images of a) 1T-PtSe₂ region; b) Janus PtSSe region. The corresponding intensity profile along c) the blue rectangle in (a); d) the yellow rectangle in (b).

ribbons. Furthermore, this uneven SHG intensity in PtS_xSe_{2-x} ribbons supports the notion that symmetry breaking in certain regions was caused by uneven doping.

Furthermore, it was noteworthy that the polarization of SHG observed in Figure 4b,c exhibited hexagonal rotational symmetry, yet the two petals demonstrated enhanced polarization along one of the armchair directions, most notably in the PtS_xSe_{2-x} sample. Considering the ribbons' armchair edge type, along with test results from three PtS_xSe_{2-x} samples with varying edge distribution densities (*D*) (Figure S8a–c, Supporting Information), the elongated polarization directions appeared to be related to the ribbons' orientation. The light spot in the experiment possessed a diameter that far exceeded the width of the ribbons (Figure S9, Supporting Information), simultaneously irradiating multiple parallel-arranged ribbons. Given the boundary effect, the ribbons lacked hexagonal symmetry relative to the incident light, implying the absence of a boundary along the ribbons' length relative to the spot and leading to irregular petal shapes protruding significantly in two directions. The polarization ratio *r* was defined as the ratio of *I_m**/*I_m*, where *I_m** represented the two maximum enhanced polarization intensities of SHG along the armchair axis, and *I_m* the other four maximum intensities along a mirror plane. Clearly, a positive correlation existed between *r*

and *D* of the ribbons (Figure S8d–i, Supporting Information). As the distribution densities increased, more ribbons contributed to the deformed polarization intensity, resulting in a higher polarization ratio along the ribbons' length compared to the other armchair directions. Monolayer PtS_xSe_{2-x} ribbons demonstrated excellent nonlinear SHG conversion efficiency, approximately twenty times greater than that of PtSe₂ ribbons, and showed strong polarization correlation. These results have inspired further exploration into the potential of monolayered PtS_xSe_{2-x} ribbons as ultra-thin frequency doubling crystals.

2.4. Electrical and Synaptic Properties of 2D PtS_xSe_{2-x} Ribbons

PtS_xSe_{2-x} (19.7% S) and undoped PtSe₂ ribbons with different layers served as channel materials (Figure S10a–f, Supporting Information) in the study of the electrical and synaptic properties through the construction of FETs. (Figure 5a). Figure 5b,c displayed typical FET transport curves of PtS_xSe_{2-x} and PtSe₂ ribbons across three distinct layers. Monolayer PtS_xSe_{2-x} and PtSe₂ ribbons each exhibited *p*-type semiconductor (*p*-SC) characteristics, which transitioned to bipolar semiconductor (BSC) characteristics with an increase to 5 layers. Upon reaching 20 or more

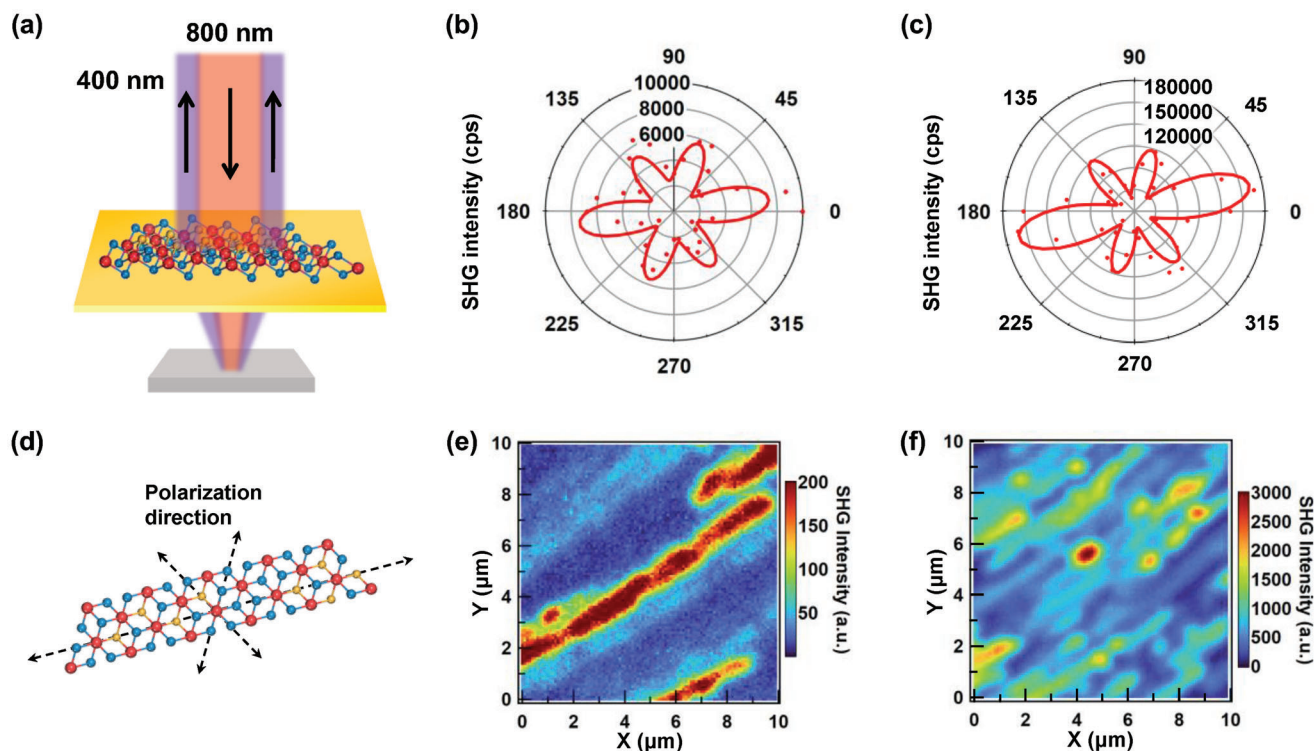


Figure 4. a) Schematic diagram of the SHG testing. Polarization-dependent SHG spectra of b) PtSe₂ ribbons; c) PtS_xSe_{2-x} ribbons. d) The six armchair directions of Janus PtSSe region. Surface scan SHG spectra of e) PtSe₂ ribbons; f) PtS_xSe_{2-x} ribbons.

layers, both types of ribbons exhibited *p*-type semi-metallic (*p*-SM) characteristics. Notably, PtS_xSe_{2-x} ribbons demonstrated improved carrier mobilities and ON-OFF ratios compared to the intrinsic PtSe₂ ribbons. In particular, for monolayer semi conductive PtS_xSe_{2-x}, the carrier mobility and the ON-OFF ratio increased by more than ten and two times, respectively. In multilayer semi metallic PtS_xSe_{2-x}, doping resulted in an ≈ 1.2 -fold increase in the carrier mobility (Figure S11, Supporting Information). The substitution of Se with S significantly altered the charge differential density (CDD) and conductivity in monolayer and few-layer PtSe₂ ribbons. Unlike the Janus structure in monolayer PtS_xSe_{2-x}, S atoms are concentrated in the lower layers of multilayer PtS_xSe_{2-x}. This difference results in varying magnitudes of carrier mobility in monolayer and multilayer PtS_xSe_{2-x}. To further elucidate the impact of S atom substitution doping on PtSe₂'s electrical properties, PtS_xSe_{2-x} crystal structures with varying doping concentrations (Figure S12a–d, Supporting Information) were modeled and analyzed using first-principles DFT calculations. Calculation results indicated that the band structure of PtS_xSe_{2-x} was similar to that of PtSe₂ (Figure S13a–d, Supporting Information), yet there were significant differences in CDD (Figure 5d). The valence band maximum (VBM) electrons in intrinsic PtSe₂ were predominantly located on Se atoms, whereas post-doping, they were distributed across Pt, S, and Se atoms. As the doping concentration increased, this phenomenon became more pronounced. Consequently, S doping reduced the localization of VBM electron distribution, enhancing the likelihood of their excitation and migration, and resulting in an increased open state current and mobility. Additionally, hystere-

sis windows were observed in the transfer characteristic curves (Figure 5b,c), more prominently in PtS_xSe_{2-x}. While intrinsic PtSe₂ possessed edge defects, PtS_xSe_{2-x} additionally exhibited a small number of in-plane Se vacancies and dangling Se atoms, which were believed to contribute to the observed hysteresis windows. Figure 5e schematically illustrated the carrier trapping-detrapping mechanism attributable to these defects. Initially, both PtSe₂ and PtS_xSe_{2-x} exhibited a type II Dirac cone band structure,^[28] with some electrons being captured in the trap state, leading to the formation of holes in the valence band. Upon application of a scanning voltage ranging from -40 to $+40$ V, electrons in the trap state were released into the channel and recombined with holes, thus reducing channel conductivity, and some of the electrons were trapped. Conversely, with a scanning voltage from $+40$ to -40 V, some channel electrons had been trapped, resulting in an abundance of holes and an increase in channel conductivity. Thus, applying a positive gate voltage pulse increased the overall channel conductivity, whereas a negative gate voltage pulse decreased it.

The PtS_xSe_{2-x}-based FETs could be utilized to simulate synaptic enhancement, inhibitory response, and repetitive learning processes in artificial synapses. The typical synaptic structure in the biological nervous system mainly comprises three parts: pre-neurons, synaptic gaps, and post-neurons (Figure 6a). Upon reaching the pre-neurons, the action potential triggers the opening of calcium channels, leading to the release of neurotransmitters into the synaptic gap, where they are received by the corresponding receptors on the post-neurons. The current resulting from ion movement across the postsynaptic membrane is

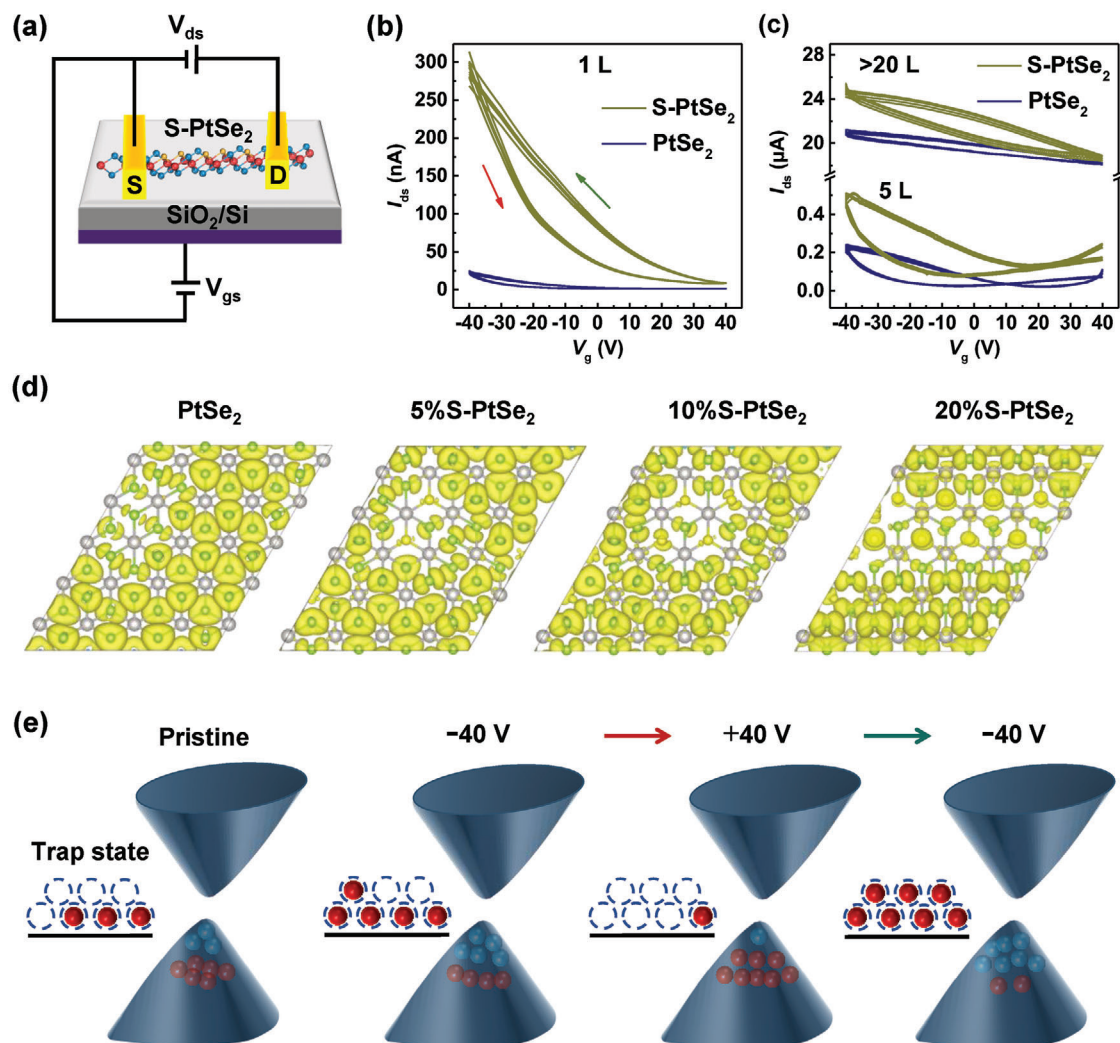


Figure 5. a) Schematic diagram of the FET device. Transport curves of $\text{PtS}_x\text{Se}_{2-x}$ and PtSe_2 ribbons with b) monolayer structure; c) multilayer structure. d) CDD of $\text{PtS}_x\text{Se}_{2-x}$ ribbons with different doping concentrations. e) Schematic diagram of the trap-detrap mechanism. The red ball, blue ball, and dotted circle represent the electrons, holes, and trap centers, respectively.

termed postsynaptic current (PSC). This process indicates that under external stimulation, action potentials can propagate from pre-neurons to post-neurons, eliciting PSC.^[29] Synaptic weight refers to the strength of connections between neurons, and its ability to adjust with neural activity is known as synaptic plasticity. Synaptic plasticity, crucial for brain learning and memory, is categorized into short-term plasticity and long-term plasticity, based on the time scales at which synaptic weights change in response to neural pulse activity. Short-term plasticity typically occurs on a time scale ranging from several tens of milliseconds to a few minutes, whereas long-term plasticity can persist for several minutes to several hours or longer. Artificial synaptic devices simulate the relationship between input pulse signals and output electrical signals of biological synapses, serving as the basis for artificial neural networks to achieve information transmission, processing, and storage.^[30] Biological synapses transmit electrical signals via ion flow across cell membranes, whereas artificial synaptic devices simulate synaptic plasticity through carrier, ion,

and proton transport in electronic materials. This work employed FET devices based on $\text{PtS}_x\text{Se}_{2-x}$ ribbons as artificial synapses, where pulse signals at V_g were input to detect changes in I_{ds} . By utilizing I_{ds} as PSC, synaptic plasticity regulation was achieved through the regulation of V_g pulse signals. Figure 6b illustrated the device's response to pulse signals. Applying a pulse signal with a width of 20 ms and an amplitude of +40 V to the device, with V_{ds} maintained at 0.1 V constant, resulted in a rapid rise followed by a gradual decline in PSC. Continuous application of two pulse signals to the device results in the PSC caused by the second pulse being significantly greater than that caused by the first, exhibiting typical paired-pulse facilitation (PPF) characteristics. The ratio of PSC amplitudes A2 to A1 caused by adjacent pulse signals is commonly defined as the PPF index. Figure 6c illustrated the time interval (Δt) between the PPF index and adjacent pulse signals. When Δt was 50 ms, the PPF index reached its maximum value of 195%. The PPF index at 50 ms is exceptional compared to other 2D synaptic devices (Figure S15, Supporting

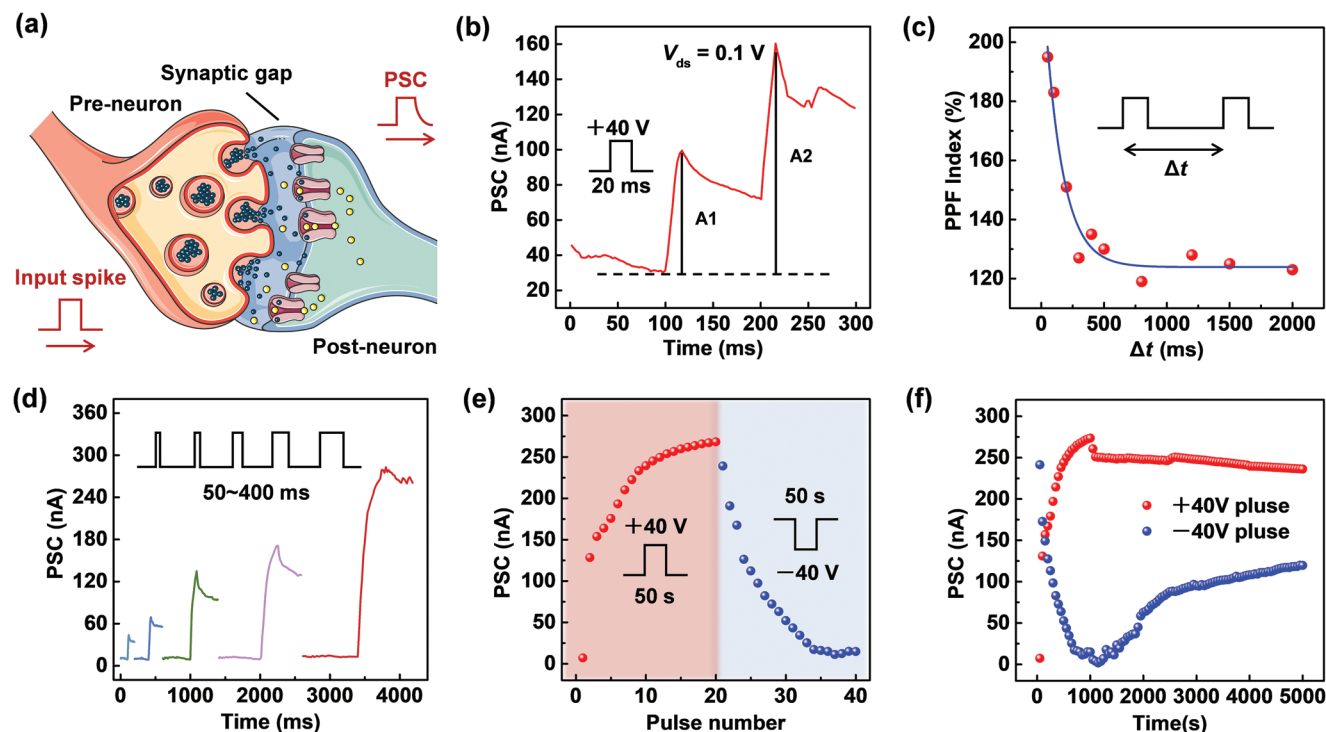


Figure 6. a) Schematic diagram of the biological synapse. b) PSC curve under two consecutive pulse excitations. c) The correspondence between PPF index and the time interval of adjacent pulse signals. d) PSC curves under different pulse widths. e) Current modulation by applying positive and negative pulses, corresponding to LTP and LTD respectively. f) Retention curves of the high-conductance and low-conductance states after LTP and LTD.

Information).^[31–40] The PPF index showed an exponential downward trend over Δt and eventually stabilizes at $\approx 125\%$. Increasing the time interval between adjacent presynaptic stimuli resulted in a decrease in the memory ability of synapses, which was consistent with the memory process of biological synapses. Figure 6d illustrated the dependence of PSC on pulse width. When the pulse width increased from 50 to 400 ms, the maximum PSC increased from 44 to 279 nA. Thus, artificial synaptic devices could generate dynamic responses to various pulse signals, providing the foundation for organisms to achieve complex neural functions. Further testing the long-term plasticity characteristics of the device. After continuously applying 20 positive and 20 negative pulses, the PSC of the device exhibited a slow decreasing and increasing trend (Figure 6e). Both excitatory and inhibitory PSCs could be maintained for over 5000 s, demonstrating good long-term potentiation and depression characteristics. Figure 6f illustrated the switching between LTP and LTD characteristics. Following the continuous application of 20 positive pulses, succeeded by 20 equally large negative pulses, the device transitions from LTP to LTD. Even after 5 cycles, the conductivity difference between the high and low states remained nearly tenfold, indicating the synaptic transistor's high repeatability (Figure S14, Supporting Information).

3. Conclusion

Overall, a method was developed to effectively achieve controllable doping of 2D PtSe₂ ribbons, in which S atoms were incorporated into the PtSe₂ lattice through substitutional doping, result-

ing in the formation of a Janus PtSSe structure. The maximum doping concentration of S atoms reached 19.7%. Compared to undoped PtSe₂, PtS_xSe_{2-x} demonstrated enhanced nonlinear optical responses, improved electrical performance, and information storage capabilities, enabling the simulation of short-term plasticity, long-term plasticity, and repetitive learning processes. Although the Janus PtSSe structure has been successfully formed in this study, preparing 2D Janus PtSSe in a large area remains challenging due to energy constraints, which implies a promising prospect for synthesizing large-scale Janus 2D TMDCs. This research offered a viable approach for investigating Janus doping modifications in 2D TMDCs and established a platform for examining the nonlinear optics of 2D TMDCs and for the design of electronic devices like artificial synaptic transistors.

4. Experimental Section

Materials Synthesis and Transfer: (NH₄)₂PtCl₄ (Alfa Aesar, 99.9% purity) aqueous solution with a concentration of 0.1–1 mg mL⁻¹, Se powder (Alfa Aesar, 99.5% purity) with a mass of 800 mg and S powder (Alfa Aesar, 99.5% purity) with a mass of 80–800 mg were used as precursors, and Au foil (Alfa Aesar, 99.99% purity, thickness ≈ 50 μ m) was the growth substrate. PtS_xSe_{2-x} and PtSe₂ ribbons were synthesized in a CVD furnace (Thermo Scientific Lindberg/Blue Series). PtS_xSe_{2-x} and PtSe₂ ribbons grown on the Au foil were transferred to a SiO₂/Si substrate by wet chemical etching. First, PtS_xSe_{2-x}/Au and PtSe₂/Au samples were coated with PMMA at 140 °C and then baked for 30 min. The samples were immersed in Au corrosives (KI + I₂ + H₂O with a mass ratio of 8:2:80) and heated in a water bath at 70 °C for 3 h to etch the Au. The PMMA-supported PtS_xSe_{2-x} and PtSe₂ were cleaned in deionized water for 3 h and then

collected on required substrates such as SiO₂/Si. Finally, the salvaged samples were heated at 70 °C for 3 h, and the PMMA was dissolved in acetone at 65 °C.

Materials Characterizations: All the samples were characterized by OM (Zeiss Axio scope A1), SEM (MERLIN VP Compact, Carl Zeiss), AFM (Cypher VRS), Raman spectroscopy (LabRAM HR Evolution, Horiba; 532 nm excitation wavelength), XPS (Escalab 250Xi, Thermo Scientific fitted with a Al K α as the radiation), STEM (FEI Titan 80–300; acceleration voltage: 300 kV) and atomic force microscope (Cypher VRS).

SHG Measurements: The excitation source was a Ti:sapphire mode-locked femtosecond laser (MaiTai SP, Spectra-Physics) generating a linearly polarized laser beam at 800 nm with a bandwidth of 30–60 nm. The 2 ω signal light generated was collected in a backscattering geometry and subsequently separated using a short-pass dichroic mirror. SHG light signals were collected in a back-scattering geometry using the same objective and detected with a Hamamatsu photomultiplier tube. A zero-order half-waveplate and a Glan-Thompson polarizer were mounted on motorized rotation stages to modify the excitation light's polarization and analyze the SHG light's polarization, respectively. When their motion steps were synchronized, a virtual sample-rotation measurement scheme was implemented, eliminating off-centering issues. A pair of galvanometer mirrors scanned the incident beam's position to capture SHG microscopy images of specific square regions.

Electrical Measurements: Using Cu meshes as masks, Au was evaporated to form the source and drain electrodes on transferred PtS_xSe_{2-x} and PtSe₂ ribbons on SiO₂/Si substrates. Subsequently, these devices were annealed at 200 °C for 30 min in a tube furnace with Ar gas serving as the protective atmosphere during the annealing process. The electrical and synaptic testing of the FET devices were performed with SourceMeters (Agilent B1500A). The carrier mobility was calculated from the slope of the transfer characteristic curve using the following formula:

$$\mu = \frac{dI_{ds}}{dV_g} \cdot \frac{L}{WC_g V_{ds}} \quad (1)$$

where μ represents the carrier mobility, $\frac{dI_{ds}}{dV_g}$ is the slope from the linear portion of the transfer characteristic curve, L and W are the channel length and width in micrometers, respectively, and C_g is the gate capacitance per unit area in nF cm⁻². V_{ds} used in the calculation was 0.1 V. The gate capacitance of the SiO₂/Si substrate used in the experiments was ≈ 3.84 nF cm⁻². The channel size of the FET device was determined using ImageJ software. The average channel size ($L \times W$) for single-layer, 5-layer, and over 20-layer nanoribbons were 15.83×1.27 , 22.32×2.61 , and 28.05×3.10 μm^2 , respectively.

Theoretical Calculations: The study used VASP to perform DFT calculations with projector augmented wave.^[41] Electron spin polarization was considered in all calculations. The Perdew–Burke–Ernzerhof functional was employed for exchange–correlation effects,^[42] while DFT+D3 was used for handling weak interactions.^[43] The cut-off energy for the plane-wave basis was 450 eV. K-points were sampled as $2 \times 2 \times 1$ grid in the Brillouin zone by means of Monkhorst–Pack method. 15 Å of layer vacuum was applied at Z-axis of slab models to avoid the Periodic effect. The transition state (TS) searches were performed using the Dimer method in the VTST package. The TS search was conducted by using the climbing-image NEB method to generate initial guess geometries, followed by the dimer method to converge to the saddle points. Energy and maximum stress were converged to 10^{-5} eV and 0.02 eV Å⁻¹, respectively.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

chemical vapor deposition, Janus, platinum diselenide, substitutional doping, synaptic transistors

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